

Rectilinear Kinematics (Ch. 12)

$v = \frac{ds}{dt}$ – velocity is the time rate of change of position

$a = \frac{dv}{dt}$ – acceleration is the time rate of change of velocity

$a ds = v dv$ – eliminates time; use when a is a function of position

$v = v_0 + a_c t$

$s = s_0 + v_0 t + \frac{1}{2} a_c t^2$

$v^2 = v_0^2 + 2a_c(s - s_0)$ – constant a_c only; keep sign convention consistent

s - t , v - t , a - t graphs:

v = slope of the s - t graph at that instant (ds/dt)

a = slope of the v - t graph at that instant (dv/dt)

Δv = area under the a - t graph (change in velocity)

Δs = area under the v - t graph (displacement)

Variable acceleration – which relation to use:

$a = f(t)$: integrate directly

$a = f(v)$: use $a = dv/dt$, separate variables

$a = f(s)$: use $a ds = v dv$, separate variables

Curvilinear & Projectile Motion (Ch. 12)

$\mathbf{v} = \frac{d\mathbf{r}}{dt} = \dot{\mathbf{r}}$ – $\mathbf{v}$ is always tangent to the path

$\mathbf{a} = \frac{d\mathbf{v}}{dt} = \ddot{\mathbf{r}}$ – $\mathbf{a}$ is generally NOT tangent to the path in curvilinear motion

$v_x = \dot{x}$, $v_y = \dot{y}$, $v_z = \dot{z}$ – rectangular velocity components; each is an independent rectilinear problem

$a_x = \ddot{x}$, $a_y = \ddot{y}$, $a_z = \ddot{z}$ – rectangular acceleration components; combine vectorially

$x = x_0 + (v_x)_0 t$

$y = y_0 + (v_y)_0 t - \frac{1}{2} g t^2 - a_x = 0$ (constant v_x), $a_y = -g$ ($g \approx 9.81 \text{ m/s}^2$)

$v_y = (v_y)_0 - g t$

$v_y^2 = (v_y)_0^2 - 2g(y - y_0)$ – solve x - and y -motion separately, linked by common time t

$v = \sqrt{v_x^2 + v_y^2}$ – resultant speed

$\theta = \tan^{-1}(v_y/v_x)$ – measured from x -axis to velocity vector; same approach for acceleration vector

Normal-Tangential (n-t) Coordinates (Ch. 12)

$v = \dot{s} \hat{u}_t$ – velocity always tangent to path, direction of increasing s

$\mathbf{a} = a_t \hat{u}_t + a_n \hat{u}_n$ – acceleration splits into tangential and normal parts

$a_t = \dot{v} = v \frac{dv}{ds}$ – tangential component; changes the *speed*

$a_n = \frac{v^2}{\rho}$ – normal component; changes *direction*; always points toward the center of curvature

$a = \sqrt{a_t^2 + a_n^2}$ – resultant magnitude of $\mathbf{a}$

$\rho = \frac{[1 + (dy/dx)^2]^{3/2}}{|d^2y/dx^2|}$ – radius of curvature, needed for a_n when path $y = f(x)$ is analytic

Circular path: $\rho = \text{const} = r$

Straight-line path: $\rho \rightarrow \infty \Rightarrow a_n = 0$, $a = a_t$ – n - t coordinates reduce to ordinary rectilinear motion

Polar & Cylindrical Coordinates (Ch. 12)

$\mathbf{r} = r \hat{u}_r$ – position

$v_r = \dot{r}$ – radial velocity component

$v_\theta = r \dot{\theta}$ – transverse velocity component

$v = \sqrt{v_r^2 + v_\theta^2}$ – always tangent to the path, as it must be

$a_r = \ddot{r} - r \dot{\theta}^2$ – radial acceleration component

$a_\theta = r \ddot{\theta} + 2\dot{r}\dot{\theta}$ – transverse; the $2\dot{r}\dot{\theta}$ term is essential whenever r varies with time

$a = \sqrt{a_r^2 + a_\theta^2}$ – resultant magnitude

$\dot{\hat{u}}_r = \dot{\theta} \hat{u}_\theta$, $\dot{\hat{u}}_\theta = -\dot{\theta} \hat{u}_r$ – unit-vector rates; source of the extra terms above

$v_z = \dot{z}$, $a_z = \ddot{z}$ – cylindrical adds the z -axis; combine vectorially:

$\mathbf{v} = v_r \hat{u}_r + v_\theta \hat{u}_\theta + v_z \hat{u}_z$

Circular path in polar form ($r = \text{const}$): $a_r = -r \dot{\theta}^2$, $a_\theta = r \ddot{\theta}$ – matches n - t with $\rho = r$

Dependent & Relative Motion (Ch. 12)

$s_A + l_{CD} + s_B = l_T$ – write a position-coordinate equation using the total constant cable length

$\dot{s}_A + \dot{s}_B = 0$ – differentiate once for velocities

$\ddot{s}_A + \ddot{s}_B = 0$ – differentiate again for accelerations; opposite signs mean the blocks move oppositely (coefficients = number of cable segments changing length at each moving pulley)

$\mathbf{r}_B = \mathbf{r}_A + \mathbf{r}_{B/A}$ – relative position; translating, non-rotating $x'y'$ frame attached to A

$\mathbf{v}_B = \mathbf{v}_A + \mathbf{v}_{B/A}$ – relative velocity

$\mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{B/A}$ – relative acceleration; for two independently-moving particles (e.g. two aircraft, a boat crossing a current) – solve as a vector equation in components

Kinetics: Equations of Motion (Ch. 13)

$\Sigma \mathbf{F} = m\mathbf{a}$ – draw a free-body diagram AND a kinetic ($m\mathbf{a}$) diagram before writing scalar equations

Rectangular components – best when path/forces are naturally in x, y, z (incline, drag/thrust problems):

$\Sigma F_x = ma_x$, $\Sigma F_y = ma_y$

$\Sigma F_z = ma_z$

Normal-tangential components – use when ρ is known (circular motion, banked curves, vertical loops):

$\Sigma F_t = ma_t = m \frac{dv}{dt}$

$\Sigma F_n = ma_n = m \frac{v^2}{\rho}$

Polar/cylindrical components – use when r, θ are the natural coordinates (e.g. rotating slotted arm):

$\Sigma F_r = m(\ddot{r} - r \dot{\theta}^2)$

$\Sigma F_\theta = m(r \ddot{\theta} + 2\dot{r}\dot{\theta})$

$\Sigma F_z = m\ddot{z}$

$\frac{d^2u}{d\theta^2} + u = -\frac{F}{mh^2u^2}$, $u = \frac{1}{r}$ – central-force/orbital motion; $F > 0$ away from force center O

$h = r^2 \dot{\theta} = \text{const}$ – angular momentum per unit mass, constant for any central force

Kinetics: Work & Energy (Ch. 14)

$U_{1-2} = \int_{s_1}^{s_2} F \cos \theta ds$ – positive when F has a component along motion

$U_W = -W\Delta y - \Delta y$ positive upward

$U_s = -\left(\frac{1}{2}ks_2^2 - \frac{1}{2}ks_1^2\right) - s$ from unstretched length; negative when stretching

$T_1 + \Sigma U_{1-2} = T_2$ - principle of work and energy (scalar); include friction as negative work

$T = \frac{1}{2}mv^2$ - kinetic energy of a particle

$P = \frac{dU}{dt} = \mathbf{F} \cdot \mathbf{v}$ - power, in watts (J/s)

$\varepsilon = \frac{P_{out}}{P_{in}} = \frac{\text{energy output}}{\text{energy input}}$ - efficiency; always < 1 (friction, heat losses)

$V_g = Wy$ - gravitational potential energy, measured from a chosen datum (+ above, - below)

$V_e = \frac{1}{2}ks^2$ - elastic potential energy; always ≥ 0 (s = deformation from unstretched length)

$T_1 + V_1 = T_2 + V_2$ - conservation of mechanical energy; valid only if all forces doing work are conservative

Kinetics: Impulse & Momentum (*Ch. 15*)

$m\mathbf{v}_1 + \Sigma \int_{t_1}^{t_2} \mathbf{F} dt = m\mathbf{v}_2$ - linear impulse-momentum; useful when $F = f(t)$

$\Sigma m_A(v_A)_1 + \Sigma m_B(v_B)_1 = \Sigma m_A(v_A)_2 + \Sigma m_B(v_B)_2$ - conservation, e.g. collisions

$e = \frac{(v_B)_2 - (v_A)_2}{(v_A)_1 - (v_B)_1} - e = 1$ perfectly elastic (KE conserved), $e = 0$ perfectly plastic (move together)

Oblique impact: resolve along line of impact (n) and plane of contact (t); momentum & e apply along n , tangential v unchanged (smooth surfaces)

$\mathbf{H}_O = \mathbf{r} \times m\mathbf{v}$ - angular momentum of a particle about O

$H_O = mvd - d =$ perpendicular distance from O to the line of action of mv

$H_O = mr^2\dot{\theta} = mrv_\theta$ - angular momentum in polar coordinates

$(H_O)_1 + \Sigma \int_{t_1}^{t_2} M_O dt = (H_O)_2$ - angular impulse-momentum; moments summed about the same fixed O

H_O is conserved (constant) whenever $\Sigma M_O = 0$, as in central-force motion

Quick Reference

$\Sigma \mathbf{F} = m\mathbf{a}$ - Newton's second law; master equation for every kinetics problem

$v = v_0 + a_c t$, $s = s_0 + v_0 t + \frac{1}{2}a_c t^2$, $v^2 = v_0^2 + 2a_c(s - s_0)$ - constant acceleration

$a_t = v dv/ds$, $a_n = v^2/\rho$ - n - t acceleration components

$a_r = \ddot{r} - r\dot{\theta}^2$, $a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta}$ - polar acceleration components

$T_1 + \Sigma U_{1-2} = T_2$ - work-energy principle

$m\mathbf{v}_1 + \Sigma \int \mathbf{F} dt = m\mathbf{v}_2$ - linear impulse-momentum

$e = \frac{(v_B)_2 - (v_A)_2}{(v_A)_1 - (v_B)_1}$ - coefficient of restitution

AER318 Dynamics — Final sheet, pages 1–2 of 3. Ch. 12–15 (identical to the midterm sheet).

Rigid-Body Planar Kinematics (Ch. 16)

Types of motion: translation (rectilinear/curvilinear, no rotation)
– rotation about a fixed axis – general plane motion (translation + rotation)

$\omega = \frac{d\theta}{dt}$, $\alpha = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2}$ – angular velocity & acceleration (fixed-axis rotation)

$\omega = \omega_0 + \alpha_c t$, $\theta = \theta_0 + \omega_0 t + \frac{1}{2}\alpha_c t^2$, $\omega^2 = \omega_0^2 + 2\alpha_c(\theta - \theta_0)$ – constant α_c

$\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$ – velocity of a point on a body rotating about a fixed axis, $\mathbf{r}$ from the axis

$a_t = \alpha r$, $a_n = \omega^2 r$ – tangential & normal acceleration of that point

$\mathbf{v}_B = \mathbf{v}_A + \boldsymbol{\omega} \times \mathbf{r}_{B/A}$ – relative velocity, translating axes; A, B on the same rigid body

$\mathbf{a}_B = \mathbf{a}_A + \boldsymbol{\alpha} \times \mathbf{r}_{B/A} - \omega^2 \mathbf{r}_{B/A}$ – relative acceleration, translating axes

Instantaneous center (IC) of zero velocity:

$v_P = \omega r_{P/IC}$ – velocity of any point P , $\perp$ to the line from the IC, proportional to distance from it

Find the IC from perpendiculars to two known velocity directions. The IC generally has nonzero acceleration – never use it for accelerations.

Relative motion, rotating axes (Coriolis): $\boldsymbol{\Omega}$ = angular velocity of the rotating xyz frame (not necessarily the body's ω)

$\mathbf{v}_B = \mathbf{v}_A + \boldsymbol{\Omega} \times \mathbf{r}_{B/A} + (\mathbf{v}_{B/A})_{xyz}$

$\mathbf{a}_B = \mathbf{a}_A + \dot{\boldsymbol{\Omega}} \times \mathbf{r}_{B/A} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}_{B/A}) + 2\boldsymbol{\Omega} \times (\mathbf{v}_{B/A})_{xyz} + (\mathbf{a}_{B/A})_{xyz}$
– the $2\boldsymbol{\Omega} \times (\mathbf{v}_{B/A})_{xyz}$ term is the Coriolis acceleration; nonzero whenever there's relative motion within the rotating frame

Rigid-Body Kinetics: Force & Acceleration (Ch. 17)

$I = \int r^2 dm$, $k = \sqrt{I/m}$ – mass moment of inertia & radius of gyration; r = distance to the axis

$I = I_G + md^2$ – parallel-axis theorem; d = distance between the parallel axes. Sum ($I_G + md^2$) for composite bodies, subtract for holes.

Moment of inertia – common shapes:

$I_G = \frac{1}{12}mL^2$ – slender rod, about center, $\perp$ to length L

$I_O = \frac{1}{3}mL^2$ – slender rod, about one end

$I_G = \frac{1}{2}mr^2$ – thin disk or solid cylinder, about central axis

$I_G = mr^2$ – thin ring/hoop, about central axis $\perp$ to plane

$I_G = \frac{2}{5}mr^2$ – solid sphere, about a diameter

$I_G = \frac{1}{12}m(a^2 + b^2)$ – rectangular plate, centroidal axis $\perp$ to plate, sides a, b

Equations of motion:

$\Sigma F_x = m(a_G)_x$, $\Sigma F_y = m(a_G)_y$, $\Sigma M_G = 0$ – pure translation (no rotation)

$\Sigma M_O = I_O\alpha$, $\Sigma F_n = m\omega^2 r_G$, $\Sigma F_t = m\alpha r_G$ – fixed-axis rotation, O on the axis; $I_O = I_G + mr_G^2$

$\Sigma F_x = m(a_G)_x$, $\Sigma F_y = m(a_G)_y$, $\Sigma M_G = I_G\alpha$ – general plane motion (3 independent scalar eqns)

Moments may also be summed about any point P : $\Sigma M_P = \Sigma M_k$, which must include the moment of $m\mathbf{a}_G$.

Rigid-Body Work & Energy (Ch. 18)

$T = \frac{1}{2}mv_G^2$ – kinetic energy, translation

$T = \frac{1}{2}I_O\omega^2$ – kinetic energy, fixed-axis rotation about O

$T = \frac{1}{2}mv_G^2 + \frac{1}{2}I_G\omega^2$ – kinetic energy, general plane motion

$U_F = \int F \cos \theta ds$ – work of a force

$U_M = \int M d\theta = M\Delta\theta$ (if M constant) – work of a couple moment

Work of weight/spring as in Ch. 14: $U_W = -W\Delta y_G$, $U_s = -\Delta(\frac{1}{2}ks^2)$, using the motion of G /spring end

$T_1 + \Sigma U_{1-2} = T_2$ – principle of work and energy, whole body or connected system; frictionless pins do no net work

$T_1 + V_1 = T_2 + V_2$ – conservation of energy; $V = V_g + V_e$ as in Ch. 14, valid only if all forces doing work are conservative

Rigid-Body Impulse & Momentum (Ch. 19)

$\mathbf{L} = m\mathbf{v}_G$ – linear momentum, acts through G

$H_G = I_G\omega$, $H_O = I_O\omega$ (fixed axis at O) – angular momentum

$H_P = I_G\omega + mv_G d$ – angular momentum about an arbitrary point P ; d = perpendicular distance from P to the line of action of mv_G

$m(\mathbf{v}_G)_1 + \Sigma \int_{t_1}^{t_2} \mathbf{F} dt = m(\mathbf{v}_G)_2$ – linear impulse-momentum, same structure as a particle but using G

$(I_G\omega)_1 + \Sigma \int_{t_1}^{t_2} M_G dt = (I_G\omega)_2$ – angular impulse-momentum; can also use I_O about a fixed point, or the IC in special cases

Conservation: linear momentum conserved with no net external impulse; angular momentum conserved about a point/axis with no net external angular impulse (e.g. impact at a pin)

Eccentric impact: combine conservation of angular momentum about the fixed point/pin (impulsive pin reactions have zero moment there) with the restitution equation $e = [(v_B)_2 - (v_A)_2]/[(v_A)_1 - (v_B)_1]$ applied to the contact-point velocities along the line of impact

Quick Reference — rigid body

$I = I_G + md^2$ – parallel-axis theorem

$\Sigma M_G = I_G\alpha$ – general plane motion (with $\Sigma F_x = ma_{Gx}$, $\Sigma F_y = ma_{Gy}$)

$T = \frac{1}{2}mv_G^2 + \frac{1}{2}I_G\omega^2$ – kinetic energy, general plane motion

$T_1 + \Sigma U_{1-2} = T_2$ – work-energy; $T_1 + V_1 = T_2 + V_2$ if conservative

$m(\mathbf{v}_G)_1 + \Sigma \int \mathbf{F} dt = m(\mathbf{v}_G)_2$, $(I_G\omega)_1 + \Sigma \int M_G dt = (I_G\omega)_2$

$v_P = \omega r_{P/IC}$ – instantaneous center method (velocities only, never accelerations)